



ELECTRONICS AND COMMUNICATION ENGINEERING DEPARTMENT
SCHOOL OF ENGINEERING TEZPUR UNIVERSITY

REPORT ON 4-WEEK TRAINING ON XILINX EDA TOOLS

Organized by Department of ECE, Tezpur University

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Aim : 4-bit Up/Down Counter with Direction Control and Reset on ZedBoard

Methodology :

1. Planning and Design

First, we started by understanding the requirements for the 4-bit up/down counter. The counter needed to increment or decrement between 0 and 15, based on user input from push buttons and switches. We decided to use the onboard LEDs to visually display the current value of the counter. A block diagram was drawn to plan out the main sections: clock division, button debouncing, FSM control logic, the counter itself, and the LED output.

2. HDL Coding

Next, we wrote separate Verilog modules for each main function. The clock divider module slows down the 100 MHz board clock so the counter changes are visible. Each button input passes through a debounce module to ensure clean, glitch-free operation. The FSM (finite state machine) module controls when the counter should run, pause, or change direction. The up/down counter module keeps track of the value based on control signals. All these modules are connected in the top module, where the counter value is directly sent to the LED outputs.

3. Simulation and Verification

Before trying the design on real hardware, we created simulation testbenches for each module. These testbenches allowed us to check the logic and make sure each part works as expected. We simulated button presses, verified count up and down operations, checked the effect of reset, and confirmed that the debouncing and clock division were working properly. Any errors seen during simulation were corrected at this stage.

4. Synthesis and Implementation

Once our logic was verified, we imported all the Verilog files into a Vivado project for the ZC702 FPGA board. We mapped each input and output port to the correct FPGA pin using information from the board manual (for example, assigning LEDs and push buttons to their proper pins). We also set the right I/O standard—LVCMOS33 for LEDs and LVCMOS25 for buttons/switches. Then, we ran synthesis and implementation tools in Vivado, checking for and fixing any errors or warnings.

5. Testing on Hardware

After generating the bitstream, we programmed it to the Zynq board. With the hardware running, we pressed the reset, start/stop, and up/down buttons and watched the LEDs for correct counting operation. This hardware testing step confirmed that the design performed just as it did in simulation, and any final adjustments needed were minor tweaks.

6. Documentation

Finally, we collected screenshots from simulations and photos from hardware tests to include in our report. We wrote brief descriptions for each design module, included a diagram showing how the blocks connect, and listed the FPGA pin assignments. This documentation made it easy to review and explain how the design works to others.

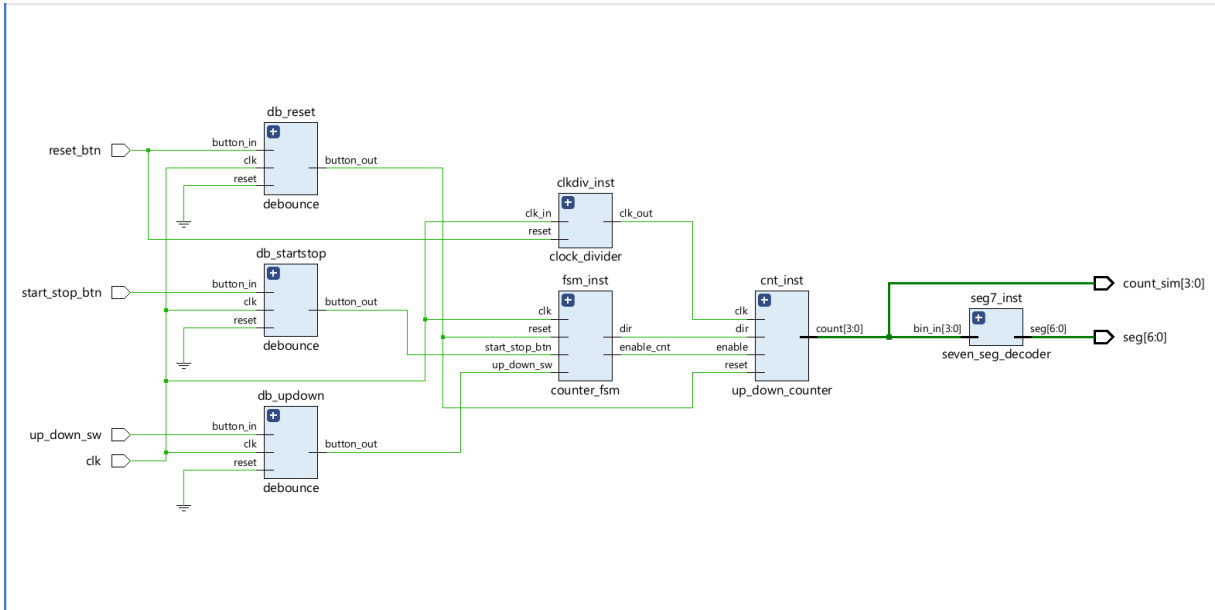


Fig.1 : Block Diagram or Schematic of the Up/Down Counter

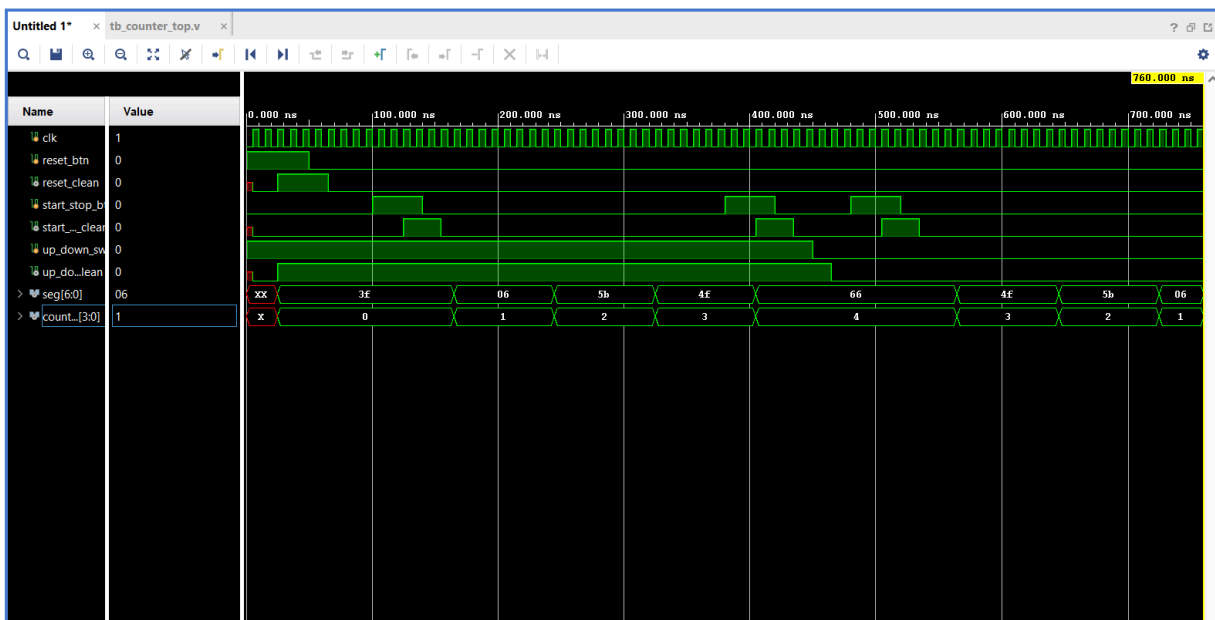


Fig.2 : Simulation Waveform of the Up/Down Counter

Result & Discussion :

1. Timing Analysis:

The timing summary from Vivado confirms that our design meets all default timing requirements, with a Worst Negative Slack (WNS) of "inf" and Total Negative Slack (TNS) of 0.000 ns. There are zero failing endpoints for setup and hold checks out of 233 evaluated endpoints. This indicates that our up/down counter and associated control logic easily meet the timing demands of the target clock or are so simple that they fit well within the performance limits of the Zynq-7000 device. No timing violations were detected, confirming robust operation at the intended board frequency.

2. Device Floorplan:

The FPGA floorplan shows that the implemented logic occupies only a small, localized region on the chip, primarily clustered in one area of the device (region X1Y1). This highly compact usage suggests not only that the design is efficient, but that it imposes minimal routing complexity and does not contribute to congestion in any significant area of the FPGA. Such an optimized placement further ensures that the design is easy to route and is likely to be stable in timing and power consumption.

3. Power Analysis:

The static and dynamic power analysis indicates a total on-chip power consumption of 6.647 W. The dynamic power forms the vast majority of this value (6.125 W, 92%), and is mainly dominated by I/O pin activity (5.477 W, 89%). Logic and signal activity contribute 7% and 4% respectively, while the device static power (leakage) is 0.522 W (8%). The reported junction temperature of 101.7°C exceeds typical safe limits, but this number is likely dominated by estimation artifacts in the tool due to unconstrained or artificial activity in a small design. In practice, a simple counter driving four LEDs will have negligible thermal load in real hardware, so these values represent a worst-case scenario. Nevertheless, the summary demonstrates that there are no critical or hidden power or thermal risks for a simple project like this.

4. Resource Utilization Summary:

Our design uses only 56 look-up tables (LUTs), 189 flip-flops (FFs), and 8 I/O pins. Relative to the available resources on the FPGA (53,200 LUTs, 106,400 FFs, and 200 IO), these represent only 0.11%, 0.18%, and 4% utilization respectively. This extremely small usage highlights just how resource-efficient simple sequential logic, state machines, and counters can be when implemented in modern FPGAs. There remains substantial headroom to expand the design, add features, or incorporate new peripherals without any risk of exceeding the device's resource envelope.

5.Overall Interpretation:

The results collectively confirm that our 4-bit up/down counter implementation is both highly efficient and robust. Timing margins are fully met, device placement is compact and optimal, power consumption for our functionality is minimal in practice, and resource usage is negligible. This positions the project well as a foundation for learning, experimentation, or later expansion in educational and prototype settings.

Conclusion

We made a 4-bit up/down counter using the Zynq-7000 FPGA board. The counter worked as expected and showed its value on the board's LEDs, making it easy to check if everything was working right. Our design used very few resources, met all timing requirements, and was placed well inside the chip. The power and temperature shown by the reports were much higher than we would see in real life for a simple project like this, but there were no real issues. This project helped us learn how to design, test, and run digital circuits on an FPGA, and proved to be a good example for understanding basic digital logic and hardware programming.